

Boyang Xu

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SUMMARY

Ph.D. candidate in Data Science developing deep generative models (flow matching, diffusion) for reinforcement learning and multi-sensor fusion, with a first-authored ICML 2026 paper and a Best Paper Award at IEEE CASE 2025. I build high-performance deep learning pipelines in JAX and PyTorch, and have hands-on experience with high-volume multi-sensor streaming data and computer vision on 3D voxel data. I enjoy creative problem solving, collaborating across disciplines, and communicating research through papers, posters, and talks.

SKILLS

- **Languages & Tools:** Python, C++, Linux/Bash, Git/GitHub, SLURM/HPC clusters
- **ML Frameworks:** PyTorch, JAX/Flax, NumPy/SciPy, Weights & Biases, TensorFlow
- **Machine Learning:** Deep Learning, Deep Generative Models (Flow Matching, Diffusion Models), Reinforcement Learning, Gaussian Processes, Bayesian Optimization, Efficient Training Pipelines
- **Perception & Sensing:** Computer Vision on 3D volumetric data (voxels), Defect Segmentation, Multi-Sensor Fusion, Time-Series Modeling, State Estimation (POMDP/HMM), Geometry & Simulation, Robotics

HONORS AND AWARDS

- Best Conference Paper Award, IEEE CASE 2025** Aug 2025
“Multi-modal Generative Modeling of Event Sequences and Time Series for Solar PV Systems”
- Finalist, Best Student Paper Competition, IISE DAIS Division** May 2023
“Data Quality Improvement and Geometric Information Recovery for Resistance Spot Welding”

PROFESSIONAL EXPERIENCE

Deep Generative Models for Distributional RL (ICML 2026) | Project Leader Tempe, AZ
Arizona State University, Graduate Research Assistant Jan 2025 – Jun 2026

- **Algorithm Design:** Designed novel flow-matching **deep generative model** algorithms (Path-Coupled Bellman Flows) for continuous-time distributional reinforcement learning; first-authored paper accepted at **ICML 2026**.
- **Deep Learning Engineering:** Engineered a high-performance **JAX-based deep learning training pipeline** from scratch with custom explicit-Euler ODE solvers, improving numerical stability and training throughput on GPU clusters.
- **Experiments & Ablations:** Designed and executed large-scale **experiments** and ablation studies across continuous-control benchmarks; shared-noise coupling and control variates **cut Wasserstein distance error by 66%**, achieving state-of-the-art results (69 on Adroit, 71 on OGBench).
- **Open Source & Communication:** Released the full framework on **GitHub** (path-coupled-bellman-flows) with reproducible scripts; communicated results through the ICML paper, poster, and research talks.

Multi-Sensor Fusion & Generative Pipeline (Best Paper, IEEE CASE 2025) Tempe, AZ
Arizona State University, Graduate Research Assistant Jan 2024 – Aug 2025

- **Sensor Fusion Pipeline:** Architected an uncertainty-aware generative pipeline fusing a **Transformer Hawkes Process** with an **iTransformer** to jointly model discrete fault events and continuous time-series signals.
- **High-Volume Streaming Data:** Processed and modeled 15 months of **high-volume multi-sensor streaming data** from an industrial solar site, fusing **34 heterogeneous sensor streams** across **19 fault types**.
- **Collaboration & Impact:** Collaborated with researchers across mechanical engineering and industrial engineering disciplines; deployed model achieved a **53.7% RMSE reduction** (0.385 → 0.178) with stable 7-day **simulation** horizons, winning **Best Conference Paper, IEEE CASE 2025**.

Autonomous UAV Active Sensing via Bayesian Optimization | Project Leader Tempe, AZ
Arizona State University, Graduate Research Assistant Sep 2023 – Aug 2024

- **Algorithm Deployment:** Programmed a cost-aware **Level-Set Bayesian Optimization** framework for active sampling and dynamic routing of autonomous UAVs performing rapid post-disaster damage assessment under resource constraints.
- **Deep Learning Integration:** Developed an **Ordinal Deep Kernel Gaussian Process** with deep neural embeddings to extract spatial features and model complex spatial **geometry** dependencies.
- **Creative Problem Solving:** Identified over **90% of critical damage areas** while cutting travel distance and compute cost by **30%** versus entropy-based baselines.

Computer Vision for 3D Industrial Inspection Tempe, AZ
Arizona State University, Graduate Research Assistant Aug 2022 – Aug 2023

- **3D Volumetric Segmentation:** Built a **computer vision** pipeline for **defect segmentation** on 3D X-ray computed tomography (XCT) **voxel** scans of metal additive manufacturing (LPBF) parts; presented at ASME MSEC 2023.
- **Spatio-Temporal Image Processing:** Developed a spatial-temporal FFT method to recover **geometric** information from thermography image sequences for resistance spot welding quality inspection; **Finalist, IISE 2023 Best Student Paper**, published in *Quality Engineering*.

EDUCATION

Arizona State University , <i>Ph.D. in Data Science (EE/CS-related STEM)</i>	Sep 2022 – May 2027 (Expected)
Northeastern University , <i>M.S. in Electrical and Computer Engineering</i>	Sep 2020 – Jun 2022
University of Jinan , <i>B.S. in Optoelectronic Information Science</i>	Sep 2016 – Jun 2020

PUBLICATIONS

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- **Boyang Xu**, Shengzhe Chen, Hao Yan. “Retimed Bellman Flows: Escaping the Impossible Triangle of Velocity Bootstrapping.” *ICLR ongoing*. Oct 2026
 - **Boyang Xu**, Mostafa Reisi Gahrooei, Mohammad Ilbeigi, Hao Yan. “Adaptive Sampling for Automated Post-Disaster Rapid Damage Assessment via Level-Set Cost-Aware Bayesian Optimization.” *SIAM SDM 2026*. Aug 2026
 - Hao Yan, Ali Sarabi, Qing Zou, **Boyang Xu**. “General Value Functions for Remaining Useful Life and Failure-Mode Prediction.” *arXiv*. Jun 2026
 - **Boyang Xu**, Qing Zou, Hao Yan. “Path-Coupled Bellman Flows for Distributional Reinforcement Learning” *ICML 2026*. May 2026
 - **Boyang Xu**, Yunyi Kang, Xinyu Zhao, Hao Yan, Feng Ju. “Partially Observable Markov Decision Process Framework for Operating Condition Optimization Using Real-Time Degradation Signals.” *Journal of Quality Technology*. Aug 2025
 - **Boyang Xu**, Shenghan Guo. “Data Quality Improvement and Geometric Information Recovery for Resistance Spot Welding with Spatial-Temporal Fast Fourier Transform.” *Quality Engineering*, 37(4), 693–705 (**Finalist, IISE 2023 Best Student Paper**). May 2025
 - Jiayu Huang, **Boyang Xu**, Yongming Liu, Hao Yan. “Multi-modal Generative Modeling of Event Sequence and Time Series for Solar PV Systems.” *IEEE CASE 2025 (Best Conference Paper Award)*. Aug 2025
 - **Boyang Xu**, Hasnaa Ouidadi, Nicholas V. Handel, Shenghan Guo. “Three-Dimensional X-ray Computed Tomography Image Segmentation and Point Cloud Reconstruction for Internal Defect Identification in Laser Powder Bed Fused Parts.” *Journal of Manufacturing Science and Engineering*, 146(9), 091002. Sep 2024
 - Shenghan Guo, Hasnaa Ouidadi, **Boyang Xu**. “Support-Structure Decision in 3D Printing with Unsupervised Computer Vision.” *ASME Int’l Manufacturing Science and Engineering Conference (MSEC)*. Jun 2024
 - Hasnaa Ouidadi, **Boyang Xu**, Shenghan Guo. “Defect Segmentation from X-ray Computed Tomography of Laser Powder Bed Fusion Parts: A Comparative Study Among Machine Learning, Deep Learning, and Statistical Image” *ASME Int’l Manufacturing Science and Engineering Conference (MSEC)*. Jun 2023

PEER REVIEW

International Conference on Machine Learning (ICML) , <i>Reviewer</i>	2026
Conference on Neural Information Processing Systems (NeurIPS) , <i>Reviewer</i>	2025
IEEE Robotics and Automation Society (RA-L / ICRA) , <i>Reviewer</i>	2025